From Interpretable to Black-Box: Dynamic Model Identification for Small USV Control Design

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***Abstract*—Effective low-level feedback control of small underactuated uncrewed surface vehicles (USVs), specifically vessels under seven meters with a single thruster and rudder, requires dynamic models that provide both actionable design insight and predictive accuracy. Tow-tank testing is impractical at this scale; parameters must be identified solely from limited field trials. Existing system identification methods span a wide range of model complexity but offer little guidance on how much fidelity is enough, because sufficiency is only meaningful relative to a stated purpose. Absent that specification, the default is a bias toward more fidelity, since the benefit of an added term is readily quantified while its costs in identifiability, experimental burden and obscured design insight are not.**

We propose a methodology built around a deliberate spectrum of three model levels, evaluated on common experimental data: (1) first-order, linear, single-input single-output (SISO) transfer functions, Nomoto-style models for surge and yaw that are immediately interpretable and support controller architecture intuition, but risk oversimplification when considered in isolation; (2) a physics-based nonlinear state-space formulation incorporating quadratic hydrodynamic drag; and (3) a Nonlinear AutoRegressive with eXogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively: simple models anchor human intuition; structured nonlinear models expose which terms are dynamically salient; and the black-box model bounds the benefit of added complexity.

Preliminary results from step-response field experiments demonstrate the approach. First-order SISO models achieve $R^2 = 0.93$ (surge, $\tau = 1.95$ s) and $R^2 = 0.83$ (yaw rate, $\tau = 0.47$ s), providing direct controller tuning parameters. Cross-model diagnostics identify quadratic drag as the dominant surge nonlinearity and reveal that speed-dependent rudder effectiveness, a key coupling for underactuated control design, is not resolvable from a single short trial, motivating targeted data collection. A margin-based diagnostic makes the sufficiency question computable, reporting the speed range over which the first-order model is sufficient for the control design. The NARX achieves one-step prediction $R^2 = 0.998$ for speed and $R^2 = 0.989$ for yaw rate on the common trial; recursive simulation on that trial falls to $R^2 = -1.988$ and $R^2 = 0.259$, respectively, and exhibits surge divergence, demonstrating that physical structure provides essential regularization under limited data. High-speed segments are retained for subsequent regime-spanning analysis.

***Index Terms*—uncrewed surface vehicle, system identification, model spectrum, control-oriented modeling, robustness margin, nonlinear dynamics, underactuated vessels**

I. Introduction

Small uncrewed surface vehicles, here taken as vessels under seven meters driven by a single thruster and rudder, are increasingly deployed for survey, inspection and autonomy research. Every one of those missions requires a working low-level controller. In production autopilots such as ArduPilot these are proportional-integral loops with an added feedforward term. Designing, simulating and tuning these control solutions is the engineering task this paper addresses.

The system identification literature offers everything from two-parameter response models to maneuvering polynomials carrying dozens of hydrodynamic coefficients. What it does not offer is guidance on which of the models is best suited for controller design. Fixing the compensator structure makes that question answerable, because it turns concrete: does a change of model level move the gains, the achievable bandwidth or the tuning procedure of that loop? A model that alters none of the three criteria, no matter how much better the fit, does not change the design.

Using the most detailed, highest fidelity, model available is not a safe default. Added model terms degrade the conditioning of the fit [1], and each new term needs excitation to identify it, which converts fidelity directly into field trials, and many model terms are unobservable via field trials. Sonnenburg and Woolsey [2] evaluate a cost function on held-out maneuvers, tabulate it per model and per regime, then make the final selection by judgment, discarding the best-scoring model because its two extra parameters are not worth the slight edge in turn-rate estimation. Skelton put the criterion in control terms in 1989, that model errors need not be small but only appropriate for the control design [3], and it has stood since without becoming computable. Making it computable for the loops above is the contribution here.

Our scope is the single thruster plus rudder vessel with field-identified parameters. Underactuated rudder steering is the interesting case, because control authority is an explicit function of forward speed, so the steering and speed channels cannot be treated as independent design problems. It is also the more representative case for the installed base of small craft, even though the USV literature is dominated by differential-thrust research platforms.

We identify three model levels from common field data, compare them with cross-model diagnostics, and give a margin-based test that reports the speed range over which added fidelity stops changing the loop design. The models buy loop structure, verification in simulation, relative comparison of candidate variants and a rehearsed tuning procedure. They do not buy gains that transfer without field tuning, and we say so plainly rather than let a reader infer otherwise.

II. Closely Related Work

The minimal control-oriented description of a steered vessel is Nomoto's first-order steering model [4], still the standard starting point in the modern formulation [5], with the identification tradition founded by Åström and Källström [6]. Physics-based maneuvering models in the Abkowitz tradition [7] carry coefficients with physical meaning but resist identification from field data, since multicollinearity among regressors leaves individual coefficients unstable even when the aggregate prediction is good [1], [8], [9].

For small craft the nearest antecedent is Sonnenburg and Woolsey [2], [10], who identify and compare a family of models for a small planing craft and evaluate them for control design, with related work spanning autonomous surface craft and small workboats [11]–[17]. The common pattern is a comparison across candidate structures followed by selection of one, after which the discarded models stop contributing.

Data-driven and grey-box marine models span recurrent architectures [18], physics-informed networks [19] and gradient-boosted ensembles [20], and our Level 3 descends from the neural identification literature of the early 1990s [21] and its delayed-input variants [22]. The distinction relevant to control is between one-step prediction, which reads measured state history at every sample, and recursive simulation, which feeds predicted states back and is what a design loop requires. Reporting practice is uneven on this point, though Woo *et al.* [18] do validate through forward simulation. Three gaps follow: sufficiency is resolved by selecting one model rather than retaining the set, evaluation is inconsistent about the one-step versus recursive distinction, and the small rudder-steered vessel is underrepresented relative to differential-thrust research platforms.

III. Model Spectrum Methodology

The three levels are not a fidelity ladder to be climbed and discarded. Each is judged by what it contributes to the PI-plus-feedforward loop of Sec. I rather than by goodness of fit: Level 1 settles the loop structure, Level 2 says where nonlinear feedforward is needed and Level 3 bounds what a black-box alternative would buy.

A. Level 1: First-Order Linear SISO Models

Level 1 treats surge and yaw as decoupled first-order channels,

$$G_v(s) = \frac{K_v}{\tau_v s + 1}, \qquad G_r(s) = \frac{K_r}{\tau_r s + 1}, \tag{1}$$

mapping normalized throttle to forward speed and normalized rudder to yaw rate. Both are identified as discrete ARX(1,1) models by ordinary least squares on the 10 Hz record and converted to continuous time. The parameters are immediately usable: τ sets the achievable loop bandwidth and K sets the loop gain, so a PI controller follows directly.

B. Level 2: Nonlinear State-Space Model

Level 2 retains the decoupled channel structure but replaces linear damping in surge with quadratic hydrodynamic drag,

$$\dot{v} = k_T\, u - d_2\, v|v|, \tag{2}$$
$$\dot{r} = k_\delta\, \delta - d_r\, r, \tag{3}$$

with u the normalized throttle and δ the normalized rudder.

Identification is two-phase, and the phasing is what makes it work. Fitting (2) jointly for k_T and d_2 is poorly conditioned, because thrust and drag are both large and nearly cancelling during steady runs, so the data constrains their difference far better than either term. Instead d_2 is estimated from coasting segments alone, where $u = 0$ and (2) reduces to $\dot{v} = -d_2 v|v|$ with a single unknown. With d_2 fixed, k_T follows from the powered segments by linear least squares.

C. Level 3: Black-Box NARX Model

Level 3 is a fixed-tap residual NARX multilayer perceptron with 19 lagged inputs, one hidden layer of 64 tanh units and two outputs which are state increments rather than absolute states. State history extends 0.5 s into the past, throttle history 2 s and rudder history 1 s. The lag structure is a project-specific implementation choice consistent with delayed-input NARX models [22]. Being multi-input multi-output, the model is free to discover the surge-yaw coupling that Levels 1 and 2 assume away.

Level 3 is a compact black-box reference point in the comparison rather than a competitor, and no claim is made that this architecture is optimal. Its role is to bound the benefit of added complexity at the available data scale. Evaluation reports one-step prediction and recursive simulation separately, following the distinction drawn in Sec. II.

D. Cross-Model Diagnostics

The value of the spectrum is the comparison across levels rather than the three fits individually. Scoring one structure on held-out data is a weaker check here, because a maneuvering model can pass a prediction test while its coefficients remain physically meaningless [8], and a coefficient that cannot be interpreted cannot inform a gain. Four diagnostics compare across structures instead. The first three ask whether a term is present in the data. The fourth asks whether it changes the loop design.

Diagnostic 1, speed-dependent rudder gain. Replace the constant k_δ in (3) with a speed-scaled $k_\delta v$ and test whether the yaw-rate fit improves.

Diagnostic 2, drag nonlinearity. Fit linear and quadratic damping to the coasting deceleration and compare.

Diagnostic 3, rudder-induced surge drag. Correlate δ^2 against the surge residual of the Level-2 fit.

Diagnostic 4, is the added fidelity inside the robustness margin? The robustness margin of a compensator is a budget for model error, so the test is whether the Level-1 to Level-2 discrepancy fits inside the margin of a controller designed from Level 1 alone. The construction has four steps.

Fig. 1. The experimental platform, a Pro Boat Blackjack 42 monohull of 1.07 m length, afloat during a field trial. Thrust comes from a single propeller and steering from a single rudder, both at the stern. The autopilot, the GNSS antenna and the radio links are mounted above the deck on the aft half of the hull.

Step 1. Design a nominal compensator from Level 1 only, in the target architecture of Sec. I, so that the test measures a change the deployed autopilot would experience rather than a property of some auxiliary controller. We use an internal model control PI, $C(s) = (\tau_v s + 1)/(K_v \lambda s)$, which cancels the Level-1 pole and gives a nominal loop $L = 1/(\lambda s)$ and complementary sensitivity $T(s) = 1/(\lambda s + 1)$, with λ the intended closed-loop time constant.

Step 2. Linearize Level 2 about operating points spanning the tested envelope. For surge, $\partial(v|v|)/\partial v = 2v_0$ for $v_0 > 0$, so (2) gives a speed-scheduled first-order plant

$$G_2(s, v_0) = \frac{K_2(v_0)}{\tau_2(v_0)s + 1}, \quad \tau_2 = \frac{1}{2d_2 v_0}, \quad K_2 = \frac{k_T}{2d_2 v_0}, \tag{4}$$

in which both the gain and the time constant fall as $1/v_0$.

Step 3. Form the multiplicative model error $\ell(\omega, v_0) = |G_2(j\omega, v_0)/G_1(j\omega) - 1|$ and test robust stability, $\|\ell T\|_\infty < 1$.

Step 4. Report the operating range over which the test passes. The output is not a yes or no but the speeds at which Level 1 stops being sufficient, which is an actionable number for a control designer and a direct answer to "enough for what."

Step 3 alone turns out to be insufficient, and Sec. V reports why. Robust stability is a statement about whether the loop remains stable, and quadratic drag is stabilizing, so at high speed the Level-1 design becomes sluggish rather than unstable and the stability test keeps passing. We therefore add a performance criterion alongside it: apply the Level-1 compensator to $G_2(s, v_0)$ and compare the achieved closed-loop bandwidth against the design intent $1/\lambda$. The pair of criteria bounds sufficiency from both sides.

IV. Experimental Platform and Data Collection

A. Vehicle

The platform, shown in Fig. 1, is a Pro Boat Blackjack 42, a production radio-controlled monohull of 1.07 m (42 in) length, instrumented to a mass of approximately 5 kg. Propulsion is single-screw: one four-pole water-cooled brushless motor turning a single propeller through a 160 A water-cooled electronic speed controller, with a single rudder driven by one digital servo and an 8S lithium-polymer pack supplying power. The vehicle is therefore underactuated in exactly the sense of Sec. I, with forward thrust on one channel and steering moment on another whose authority depends on forward speed.

Choosing a production hull rather than a purpose-built platform is deliberate. Vessels of this class are what the small-USV installed base looks like, and the identification problem is posed by the vehicle that exists rather than by one designed to be identifiable.

B. Autopilot, Sensing and Logging

Guidance, control and logging run on a Cube Orange autopilot executing ArduRover 4.6.3 (commit `3fc7011a`), with Mission Planner as the ground station. Position and ground speed come from a HERE3 GNSS receiver over DroneCAN, and angular rates from the autopilot's internal inertial measurement unit.

The actuator configuration is confirmed by the parameter set: `SERVO1` is assigned to ground steering and `SERVO3` to throttle, and every other servo output is disabled. There is one thrust input and one steering input, with no differential-thrust path. Commands are logged as `RCOU_C3` (throttle) and `RCOU_C1` (rudder) at 1100–1900 μs PWM and normalized to $[-1, 1]$ about the 1500 μs neutral point. Measured outputs are `GPS_Spd` ground speed at 5 Hz and `IMU_GyrZ` yaw rate at 137 Hz. The trials used here are flown in manual mode, so the logged commands are open-loop operator inputs rather than the output of any autopilot loop.

The autopilot's stabilization layer is the design target named in Sec. I, and it is worth stating concretely because it is what the model levels are ultimately for. Two independent loops run at that layer: throttle to forward speed (`ATC_SPEED_*`) and rudder to yaw rate (`ATC_STR_RAT_*`), each nominally a PID with an added feedforward term. On this vehicle the derivative gain of both loops is set to zero, so the deployed compensator is proportional-integral plus feedforward, matching the structure assumed in Sec. III. The remaining gains are $P = 0.4$, $I = 0.1$, $\mathrm{FF} = 0.2$ on the speed loop and $P = 1.0$, $I = 0.77$, $\mathrm{FF} = 0.2$ on the yaw-rate loop.

The configured cruise speed for the vehicle is 5 m/s. The identification trial described below spans 0–2.9 m/s, so it covers roughly the lower half of the speed range the vehicle is intended to operate over. This bears on Diagnostic 4 in Sec. V-D, whose sufficiency interval is narrower still.

C. Data Cleaning

Multi-rate streams are aligned to a common 10 Hz grid by linear interpolation. Records are segmented at mode changes, signal-freshness discontinuities and quality flags, so that no segment spans a transition that the models are not intended to represent.

D. Datasets

Levels 1 and 2 are identified from a single step-response trial, a 130 s working window drawn from a 341 s recording, covering throttle 0–39% and speed 0–2.9 m/s and comprising 1,300 samples at 10 Hz. One 29.3 s gap caused by vessel repositioning is handled by segmentation. This is intentionally minimal, a proof of concept for the workflow rather than a characterization of the vehicle.

Level 3 draws on a larger corpus: 44,133 samples in 70 slow-region segments from 65 source logs. The slow-region rule excludes a complete segment once speed stays above 3.4028 m/s for at least 1 s continuously. Allocation is grouped by source log, so every segment from one recording falls in a single partition and no recording is split across training and validation. Training uses 37 logs, being 40 segments and 28,023 rows. Validation uses every other viable slow-region recording, being 28 logs, 30 segments and 16,110 rows. Ten complete high-speed segments consistent with planing are held aside for later regime-spanning work and are not used in the results below. That label is empirical GPS speed over ground rather than a verified physical planing transition.

For direct comparison across levels, the frozen Level 3 model is also evaluated on the same 130 s trial used for Levels 1 and 2. Its 20-sample lag history leaves 1,279 scored rows against the 1,300 rows available to Levels 1 and 2.

V. Results

A. Level 1

Ordinary least squares on the step-response trial gives

$$G_v(s) = \frac{6.42}{1.95s + 1}, \qquad G_r(s) = \frac{0.771}{0.471s + 1}, \tag{5}$$

with simulation $R^2 = 0.930$ in surge and 0.826 in yaw rate. The surge time constant of 1.95 s and the yaw time constant of 0.471 s differ by a factor of four, which alone settles the loop structure: the two channels can be given separate loops at separate bandwidths.

B. Level 2

Two-phase identification gives

$$\dot{v} = 3.963\, u - 0.438\, v|v|, \tag{6}$$
$$\dot{r} = 1.474\, \delta - 1.912\, r, \tag{7}$$

with $R^2 = 0.909$ in surge and 0.820 in yaw rate. Both are marginally below Level 1 on this trial, which is expected: Level 1 is fit to minimize exactly this error on exactly this data, while Level 2 estimates its drag coefficient from coasting segments alone. The useful comparison is what the structure exposes rather than the R^2, and Sec. V-D takes that up.

Estimating d_2 from coasting is what makes the fit usable. The joint fit is poorly conditioned for the reason given in Sec. III, and the resulting parameters are sensitive to the powered-segment weighting in a way the coasting estimate is not.

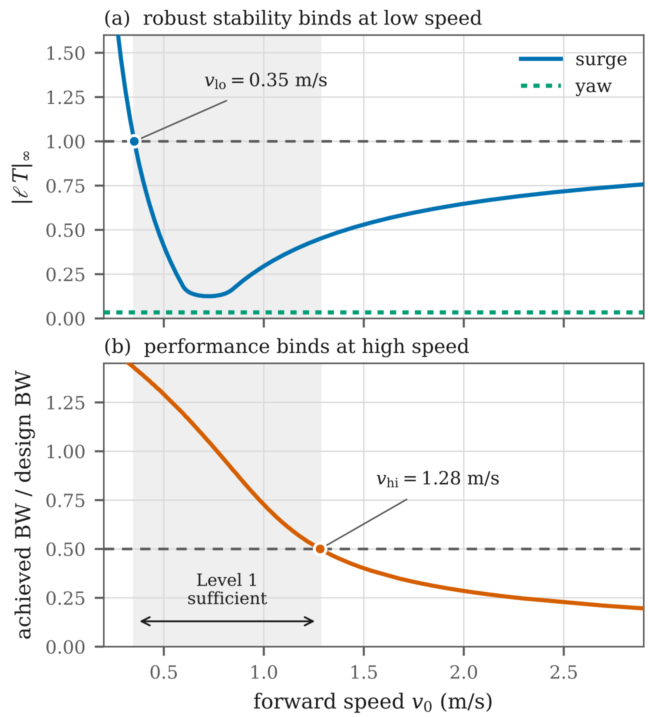

Fig. 2. Diagnostic 4. (a) Robust stability of the Level-1 compensator against the Level-2 linearization binds at low speed, where the quadratic-drag linearization is singular. (b) Achieved closed-loop bandwidth binds at high speed. The shaded band is the interval over which the Level-1 model supports the design. The yaw channel (a, dotted) never approaches the limit, so Level 2 changes nothing for yaw control.

C. Level 3

On the full slow-region validation set of 30 segments from 28 held-out logs, the one-step macro-NRMSE is 0.084 and the recursive macro-NRMSE is 0.425. Speed R^2 falls from 0.998 one-step to 0.836 recursive, and yaw-rate R^2 from 0.991 to 0.768. Recursive simulation covers every validation sample. Fig. 3 shows the surge channel across that set.

The comparison on the common trial is sharper. Level 3 reaches one-step R^2 of 0.998 in speed and 0.989 in yaw rate, above both structured models on the same data. Under recursive simulation on that same trial it falls to -1.988 in speed and 0.259 in yaw rate. A negative R^2 means the model is a worse predictor than the mean of the measured signal. Fig. 4 shows what that looks like: the surge trajectory drifts below zero and stays there for tens of seconds while Levels 1 and 2 track the measured response.

The gap between one-step and recursive is the result. One-step prediction reads the measured state at every sample, so a model can score 0.998 while having learned little more than that the state changes slowly. Recursive simulation removes that support, and the small per-step increment errors accumulate without any physical constraint to arrest them. Levels 1 and 2 cannot drift this way because the drag term bounds the surge state structurally.

D. Cross-Model Diagnostic Summary

Diagnostic 1 finds no improvement from a speed-dependent rudder gain ($\Delta R^2 = -0.03$). The trial holds forward speed nearly constant during the rudder steps, so the regressor is barely excited. This flags a gap in data coverage rather than absence of the effect, and it is the primary target of the next collection campaign.

Diagnostic 2 finds quadratic damping clearly dominant, with $\Delta R^2 = +0.20$ over the linear fit, $d_1 \approx 0$ and $d_2 = 0.438$ m^{-1}. This is the direct empirical warrant for the Level-2 structure.

Diagnostic 3 finds rudder-induced surge drag negligible, with a correlation of $r = -0.08$ between δ^2 and the surge residual. The coupling term is not warranted by this data.

Diagnostic 4 is reported in Fig. 2, computed with $\lambda = 1.0$ s. The result is an interval rather than a threshold. Robust stability fails below $v_0 = 0.35$ m/s, and this lower bound has the closed form

$$v_{\text{lo}} = \frac{k_T}{4d_2K_v}, \tag{8}$$

which is independent of λ because $|T| \leq 1$ places the supremum of $\ell|T|$ at DC. The mechanism is the $1/v_0$ singularity in (4): quadratic drag provides no linear damping at zero speed, so the linearized gain and time constant both diverge and no first-order fit can track them.

At the upper end the stability test never fails. At 2.9 m/s the model error is still within budget at $\|\ell T\|_\infty = 0.76$. What fails is performance: the Level-1 compensator applied to the Level-2 plant retains only 20% of its intended bandwidth at 2.9 m/s, crossing 50% at $v_{\text{hi}} = 1.28$ m/s. Quadratic drag is stabilizing, so the Level-1 design at speed is conservative rather than unsafe. The failure mode is sluggishness.

Taken together, the Level-1 surge model supports the control design only over roughly $[0.35, 1.28]$ m/s, under half of the tested envelope. Both bounds are insensitive to the design knob: sweeping λ from 0.5 to 3 s moves v_{hi} only between 1.23 and 1.61 m/s and leaves v_{lo} unchanged.

For yaw the picture is different. Level 2 linearizes to $r/\delta = 0.771/(0.523s + 1)$ against Level 1's $0.771/(0.471s + 1)$: identical DC gain and an 11% difference in time constant, giving $\|\ell T\|_\infty = 0.034$. Level 2 buys nothing for yaw control across the whole envelope, and we say so rather than let the added structure imply otherwise.

VI. Discussion

A. What the Spectrum Reveals

For this vehicle class and at the speeds tested, surge and yaw are weakly coupled, so decoupled SISO controllers are a defensible starting point. Quadratic drag is significant, which means a speed-hold controller needs nonlinear feedforward and a purely linear controller will show speed-dependent steady-state error. Whether the rudder gain scales with forward speed can be neither confirmed nor denied from this data, and that is the single largest gap the expanded dataset must fill.

No one of the three models exposes all of this. Level 1 alone gives no reason to suspect the drag is quadratic. Level 2 alone does not reveal that its extra structure is inert for yaw. Level 3 alone would report an excellent one-step score and mislead anyone who took it to simulation.

B. Implications for Compensator Architecture

Level 1 supports a baseline PI structure with directly identified gains, valid over the interval Diagnostic 4 returns. Level 2 justifies nonlinear feedforward on the surge channel, and Diagnostic 4 says at what speed it becomes necessary; on the yaw channel it changes nothing, so the added structure should not be carried into the design. Gain scheduling on the rudder loop stays open, gated on Diagnostic 1. Level 3 bounds what lies beyond the fixed architecture: a black-box model free to discover the coupling the structured levels assume away is the natural precursor to model-predictive or learning-based control, and its recursive failure says plainly that the data does not yet support moving past PI plus feedforward. None of the three, though, delivers `ATC_SPEED` and `ATC_STR_RAT` gains that work on the water without adjustment. What the spectrum buys is the loop structure, a simulation to verify it in, a basis for comparing variants and a tuning session that can be planned rather than improvised.

C. Predictive Fit Against Simulation Usefulness

The Level-3 result is the paper's sharpest empirical claim. A model can be the most accurate one-step predictor in the comparison and simultaneously the least useful simulator, and the two properties are measured by numbers that differ by more than two units of R^2 on identical data. Any evaluation that reports only one-step accuracy is silent about the property a control designer needs, which is whether the model can stand in for the vehicle in a closed loop.

The obvious objection is that the black-box model simply needs more data. That objection is weaker than it looks here. Level 3 was trained on 28,023 samples against the 1,300 used to identify Levels 1 and 2, roughly twenty times as many, and still produced the least usable simulator of the three. More data would presumably narrow the gap, and the sensitivity of each level to training-set size deserves direct measurement, but at the volumes a small-boat program can realistically collect the structured models are ahead. Physical structure is useful here because it constrains what the model can do once the measurements are taken away, and that constraint is worth more than samples at this scale.

D. Limitations

Levels 1 and 2 rest on a single trial covering 0–2.9 m/s and throttle to 39%, so the identified parameters are not a characterization of the vehicle across its envelope. Diagnostic 4 inherits that limitation, since the Level-2 linearization it uses is only as good as the parameters behind it. The common-trial comparison in Sec. V-C rests on one 130 s record, which is enough to demonstrate the divergence but not to quantify how often it occurs. Level 3's training corpus is broader but restricted to the slow region by construction, so nothing here speaks to behavior near or above the planing transition. Since

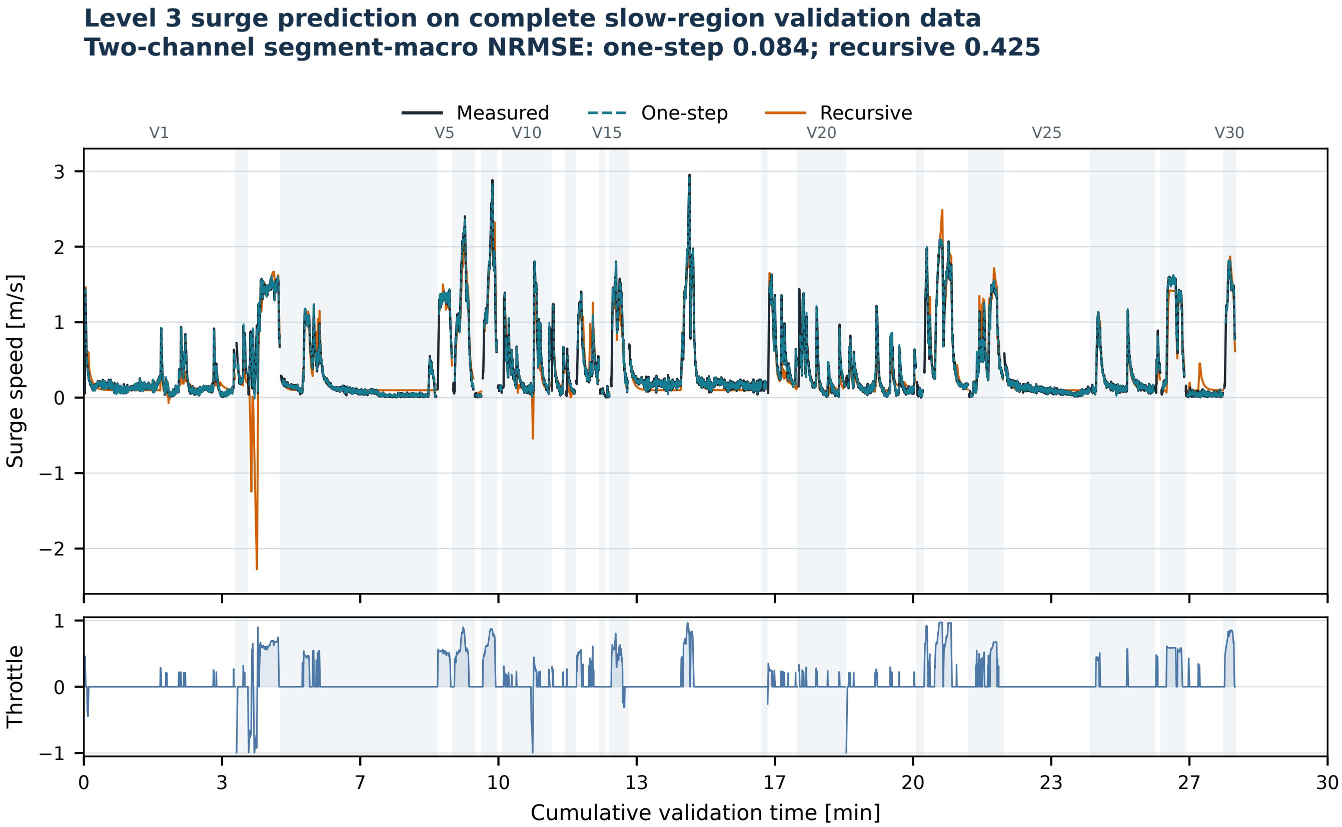

Fig. 3. Level 3 surge prediction on the complete slow-region validation set of 30 segments from 28 held-out source logs. Each segment is initialized with 2 s of measured history. One-step prediction continues to use measured state history; recursive simulation feeds predicted states back while replaying recorded throttle. Alternating backgrounds and short gaps mark segment boundaries. Reported NRMSE values are segment-macro averages over speed and yaw-rate errors normalized by training-only channel scales.

the configured cruise speed is 5 m/s, none of the three levels is characterized over the upper half of the vehicle's intended operating range. All results come from a single vehicle, so generalization across hull form and scale within this class is untested here.

VII. Conclusion

We have presented a three-level model spectrum for small underactuated USVs, identified from field trials and evaluated on common data, and posed throughout against a single concrete question: what does a practitioner need in order to design, simulate and tune the speed and yaw-rate loops of a production autopilot? The central finding is that the spectrum collectively answers that question better than any single model does. Simple models are a persistent tool for this work rather than a stepping stone to be discarded once a better fit is available.

The margin-based sufficiency test converts Skelton's qualitative criterion into a computed interval for this vehicle class. For the surge channel it returns $[0.35, 1.28]$ m/s, bounded below by robust stability and above by loss of closed-loop bandwidth, and it reports that the same test is satisfied everywhere on the yaw channel, where the added Level-2 structure is inert.

Future work runs in three directions. The first is data: multiple trials spanning the full throttle range and dedicated rudder sweeps at fixed speeds, which is what resolves the speed-dependent rudder gain that Diagnostic 1 could not settle. The second is a stronger basis for cross-level comparison, since the present common trial is a single 130 s record and a multi-trial evaluation set would let the Level-3 divergence be quantified rather than demonstrated. The sensitivity of each level to training-set size belongs here too, particularly across displacement, semi-planing and planing regimes. The third is to treat the robustness margin as a formal sufficiency criterion across the spectrum, computing for each design decision the level at which further fidelity is provably inert, which would replace the expert judgment of Sec. II with a computable bound on how much model a given autopilot loop requires.

The dataset and analysis notebooks will be released.

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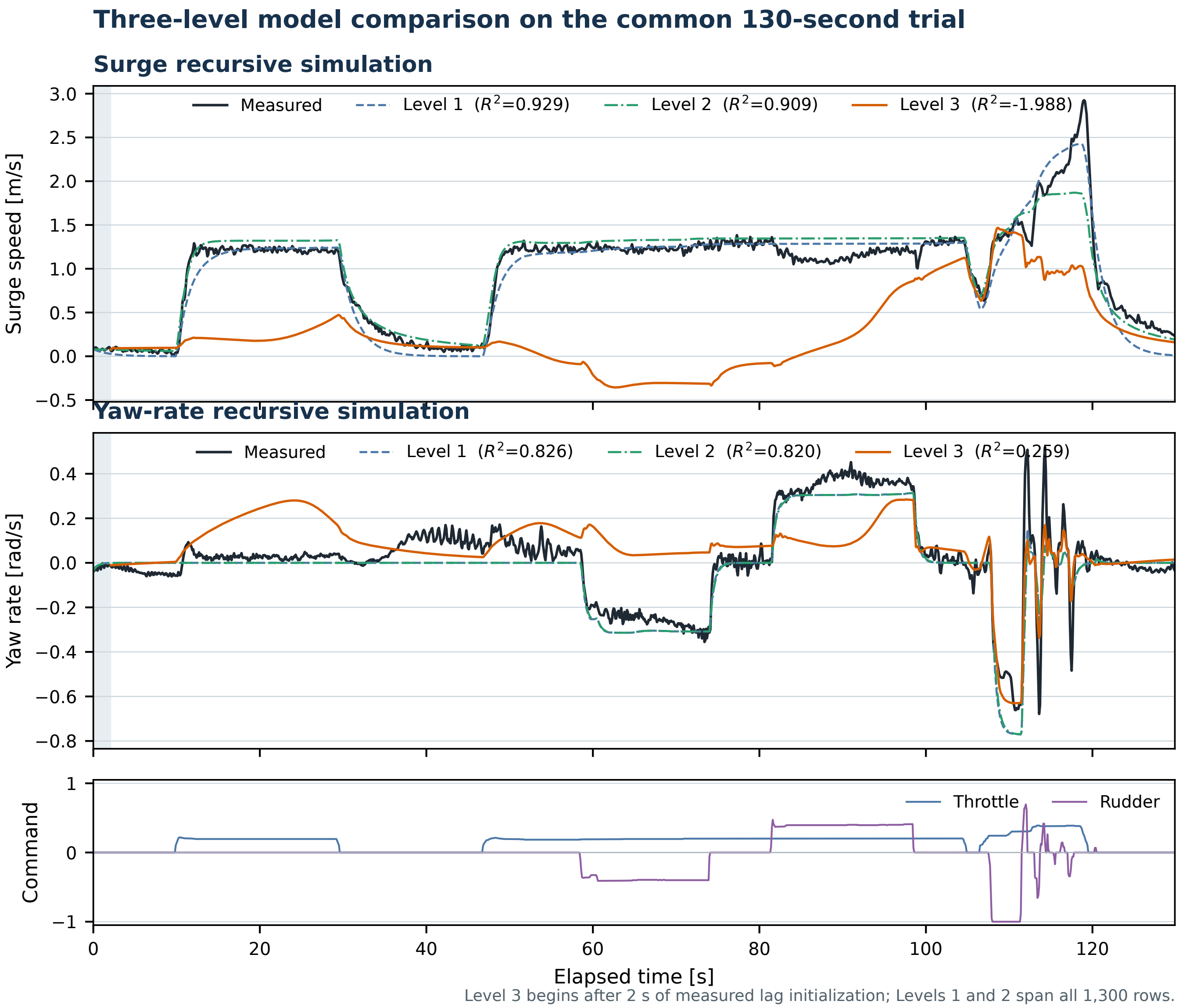

Fig. 4. All three levels under recursive simulation on the common 130 s trial. Levels 1 and 2 track the measured response; Level 3 diverges in surge, driving the predicted speed below zero for tens of seconds despite reaching the highest one-step accuracy of the three. Level 3 begins after 2 s of measured lag initialization, so it scores 1,279 rows against 1,300 for Levels 1 and 2.

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